Beyond the KV Cache: Why Matrix Compression Blurs the Past and Local Halos Preserve It

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Abstract

As transformer context windows scale to hundreds of thousands of tokens, the linear growth of the Key-Value (KV) cache becomes the primary bottleneck in inference memory and latency. Efforts to replace the growing KV cache fall into two main categories: recurrent state-space or linear attention models that compress history into a fixed-size matrix, and selective token-eviction methods that prune the cache to a small subset of discrete entries. Through controlled interventions on pretrained foundation models, we characterize the structural trade-offs governing both paradigms. First, we show that retrofitting softmax attention into a fixed-size recurrent matrix state—after removing Rotary Position Embedding (RoPE) phase interference via a complex conjugate transform—recovers over 95% of full-attention output variance and preserves general language modeling and code generation. However, because a fixed-rank state acts as a spectral low-pass filter whose signal-to-noise ratio decays inversely with sequence length, matrix compression fails on exact single-token retrieval once target facts pass outside the local sliding window. Conversely, standard token-eviction policies preserve discrete representations but struggle when a query references a distant fact by its surrounding semantic context rather than its exact surface tokens, or when the target answer spans multiple tokens. We observe that retrieval queries induce a localized attention halo over the clauses flanking a target fact, and that widening the salience pooling radius around recent observation queries captures both the semantic anchor and the complete multi-token value span. Evaluated across models from three organizations (`Qwen3.5-0.8B`, `Llama-3.2-1B`, and `SmolLM2-1.7B`) alongside the RULER and LongBench suites, wide-radius halo selection preserves retrieval accuracy and complex document question answering while clamping the active KV cache to a fixed token budget, enabling up to 8× higher multi-tenant batch concurrency at 32k context length.

1 Introduction: The Memory Bottleneck and Alternatives to the KV Cache

Autoregressive Transformers rely on the Key-Value (KV) cache to store the intermediate representations of every preceding token $(\mathbf{k}_t, \mathbf{v}_t) \in \mathbb{R}^{d_k} \times \mathbb{R}^{d_v}$. While caching prevents redundant computation during generation, its memory footprint and attention bandwidth scale linearly with sequence length $\mathcal{O}(T)$. At long contexts ($T \geq 32$k) or high multi-tenant batch sizes, reading the KV cache from high-bandwidth memory (HBM) dominates decoding latency and exhausts GPU capacity.

Efforts to bound or eliminate the growing KV cache follow two distinct architectural philosophies:

1.1 Matrix-State Compression (Linear Attention, SSMs, and Test-Time Regression)

The first paradigm replaces the expanding token table $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{T \times d}$ with a fixed-size associative memory matrix $\mathbf{S}_t \in \mathbb{R}^{d_\phi \times d_v}$ updated recurrently in $\mathcal{O}(1)$ space per step:

$$\mathbf{S}_t = \gamma_t \mathbf{S}_{t-1} + \phi(\mathbf{k}_t)\mathbf{v}_t^\top, \qquad \mathbf{y}_t = \mathbf{S}_t^\top \phi(\mathbf{q}_t) \tag{1}$$

This formulation unifies Linear Attention (Katharopoulos et al., 2020; Choromanski et al., 2021), structured State-Space Models such as Mamba and Mamba-2 (Gu & Dao, 2023; Dao & Gu, 2024), Gated DeltaNet (Yang et al., 2024), RWKV (Peng et al., 2023), and Test-Time Training (TTT) layers (Sun et al., 2024). Rather than training from scratch, recent retrofit methods such as LoLCATs (Zhang et al., 2024) and SUPRA (Mercat et al., 2024) attempt to distill pretrained softmax attention layers directly into linear recurrent states. More expressive variants view the state matrix as an online regression solver—updating $\mathbf{S}_t$ via Recursive Least Squares (RLS) or fast-weight delta rules so that $\mathbf{S}_t^\top \phi(\mathbf{k}_i) \approx \mathbf{v}_i$ across historical tokens.

1.2 Discrete KV Cache Eviction and Sparse Selection

The second paradigm retains exact discrete token representations $(\mathbf{k}_i, \mathbf{v}_i)$ but bounds the cache size to a fixed budget $M \ll T$ by evicting uninformative tokens:

- **Windowed Attention and Attention Sinks (StreamingLLM):** Xiao et al. (2024) showed that retaining the first few initial tokens ("attention sinks") alongside a sliding window W stabilizes perplexity over infinite horizons, though any information falling outside the window is permanently discarded.
- **Cumulative Attention Eviction ($\mathbf{H_2O}$ and Scissorhands):** Zhang et al. (2023) and Liu et al. (2023) accumulate historical attention scores across all decoding steps to retain "Heavy Hitter" tokens. Because early structural tokens accumulate weight over thousands of steps, cumulative scoring suffers from severe length bias that crowds out newly introduced facts.
- **Observation-Window Pooling (SnapKV):** Li et al. (2024) observed that tokens relevant to a user's prompt receive heightened attention from the final W_{obs} tokens of the prompt itself. By applying a narrow 1D pooling kernel (typically kernel size $K_{\text{pool}} \in \{3, 5\}$, radius $K \in \{1, 2\}$) over prompt-tail attention weights, SnapKV selects discrete KV clusters without cumulative length bias. Related indexers such as Quest (Tang et al., 2024) and NSA (Yuan et al., 2025) group keys into contiguous blocks for page-level selection.

1.3 Core Questions and Contributions

Despite rapid progress along both tracks, a fundamental architectural question remains unresolved: When should an inference engine compress history into a continuous matrix state, and when must it preserve discrete tokens? Furthermore, within discrete token selection, why do existing observation-window policies fail when retrieving multi-token values or facts referenced by indirect semantic context?

In this work, we investigate both paradigms through controlled architectural interventions and diagnostic benchmarks:

1. **Why Matrix States Blur the Past (Section 2):** We show that retrofitting softmax attention into a continuous ridge-regression state faces two obstacles: local positional routing collapse and Rotary Position Embedding (RoPE) phase interference. While combining a local sliding window with a phase-invariant complex conjugate lift (**PRISM**, Phase-Rotated

Invariant Spectral Mapping) recovers over 95% of full-attention output variance and preserves general language modeling and code generation at high compression, any fixed-rank matrix state acts as a spectral low-pass filter whose per-token signal decays as $\mathcal{O}(1/T)$. Consequently, matrix states preserve global topical gist but fail on exact multi-digit passkey retrieval beyond the local window.

2. **The Salience Halo Principle (Section 3):** Turning to discrete token selection, we identify why standard point-salience eviction and narrow observation-window pooling break down under realistic retrieval conditions. When a query references an earlier entity by its surrounding clause rather than its exact surface tokens, or when the target answer spans multiple tokens, narrow pooling truncates the value sequence. Because natural language facts are embedded in local syntactic clauses, query attention forms a broad salience halo around the target tokens. By widening the salience diffusion kernel ($K = 16$) and pairing it with phase-invariant retrieval scoring (**HALO**), we capture both the semantic anchor and the complete multi-token answer span.

3. **Empirical Generalization and Serving Systems (Sections 4 & 5):** Across standardized long-context benchmarks and three distinct foundation model families, wide-radius halo selection preserves retrieval accuracy where narrow pooling degrades. In multi-tenant serving at extended contexts, clamping active KV history to local halos eliminates memory-bandwidth bottlenecks, shifting generation into compute-bound scaling and expanding batch concurrency by an order of magnitude.

2 Why Matrix-State Compression Blurs the Past

To understand the limits of replacing the KV cache with a fixed-size matrix, we examine a hybrid foundation model (`Qwen3.5-0.8B`, comprising 18 linear Gated DeltaNet layers interleaved with 6 full Softmax attention layers) and replace its remaining full-attention KV caches with an online ridge-regression state $\mathbf{S}_t = (\mathbf{\Phi}_t^\top \mathbf{\Phi}_t + \lambda \mathbf{I})^{-1} \mathbf{\Phi}_t^\top \mathbf{V}_t$ (**BLUR**, Background Least-Squares Unselective Recurrence).

2.1 The Need for a Local Window and Phase De-Rotation

Directly substituting a global regression matrix for a pretrained softmax layer causes immediate generation collapse. Pretrained attention heads perform two distinct operations simultaneously: sharp, position-sensitive syntactic routing over adjacent tokens, and broad semantic aggregation over distant history. A fixed-rank matrix state cannot replicate sharp local routing without exhausting its spectral degrees of freedom. Retaining a local sliding window for recent tokens while routing evicted history into the regression state restores language modeling fluency and reasoning capacity, compressing distant context by nearly a thousand-fold.

Even with a local window, fitting a linear kernel directly to post-RoPE keys $\tilde{\mathbf{k}}_t = \mathbf{R}_{\Theta,t}\mathbf{k}_t$ recovers less than half of the native attention output variance ($R^2 = 0.4014$). Because RoPE multiplies each two-dimensional subspace by a high-frequency rotation $\theta_{t,m} = t\omega_m$, identical semantic tokens appearing at different positions point in rapidly oscillating directions, causing destructive cancellation when summed into $\mathbf{\Phi}^\top \mathbf{V}$. Applying the inverse rotation $\mathbf{R}_{\Theta,-t}$ prior to feature lifting recovers content-stationary keys, while a complex conjugate polynomial expansion (**PRISM**, Phase-Rotated Invariant Spectral Mapping: $\psi_m(\tilde{\mathbf{k}}_t) = [\tilde{k}_{t,2m} + i\tilde{k}_{t,2m+1}]^2 e^{-i2\theta_{t,m}}$) preserves relative positional selectivity alongside linear state additivity. Closed-form Ridge regression over PRISM features recovers a mean $R^2 = 0.9505$ across all softmax layers and matches the full model's zero-shot code generation accuracy on HumanEval (40.0% pass@1).

2.2 The Spectral Low-Pass Filter Limit

Despite recovering over 95% of attention variance and preserving fluency and code generation, every continuous matrix state variant scores 0.0% on exact multi-digit passkey retrieval once the target fact falls outside the local sliding window ($T - t_{\text{needle}} > W$).

This failure is not an optimization artifact; it is an information-theoretic property of fixed-rank linear projections. A regression state $\mathbf{S}_t \in \mathbb{R}^{r \times d_v}$ stores a superposition of T historical key-value outer products across r orthogonal basis dimensions. By the Eckart-Young-Mirsky theorem, rank-r projection preserves the top r principal components of the context—capturing global topical themes and recurring syntactic patterns that reinforce across hundreds of tokens—while attenuating isolated, high-frequency directions. A single passkey or UUID occurs at only $k \ll r$ token positions; its contribution to the covariance matrix $\mathbf{\Phi}^\top \mathbf{\Phi}$ scales as $\mathcal{O}(k/T)$, which is drowned out by the $\mathcal{O}(1)$ variance of the surrounding haystack (Figure 1, Panel B). Matrix states act as spectral low-pass filters: they compress the global gist of a document, but blur isolated needles.

3 Preserving the Needle: The Salience Halo Principle

If exact retrieval requires discrete token representations, the inference engine must maintain a bounded cache of $N \ll T$ discrete KV pairs selected from history. However, standard eviction policies fail whenever the query does not share exact surface tokens with the target value.

3.1 Why Point-Salience and Narrow Pooling Fail

Consider two common retrieval patterns in natural language:

1. **Retroactive Contextual Retrieval (Semantic-to-Value Separation $\Delta_{\mathbf{QA}}$):** A document introduces a fact within a clause—e.g., *"During the **initial deployment phase**, the system was calibrated under protocol **84920**"*—and the user later asks, *"Which protocol was used during the **initial deployment phase**?"* The query shares semantic features with the context words (*"initial deployment phase"*), which sit Δ_{QA} tokens away from the numeric value (*"84920"*). Point-salience methods that score individual tokens in isolation select the matching context words but evict the adjacent number they modify.

2. **Multi-Token Value Spans ($L_{\mathbf{val}} > 1$):** When the target fact is a compound identifier, date, or multi-part UUID (`"994-ALPHA-772-OMEGA"`), the model's query vector typically attends strongly to the leading token or structural delimiter before autoregressively reading the remaining tokens. Narrow pooling windows (such as SnapKV's default radius $K \in \{1, 2\}$) capture the leading sub-token but truncate the tail of the span, causing Exact Match accuracy to drop to zero even when the entity is located.

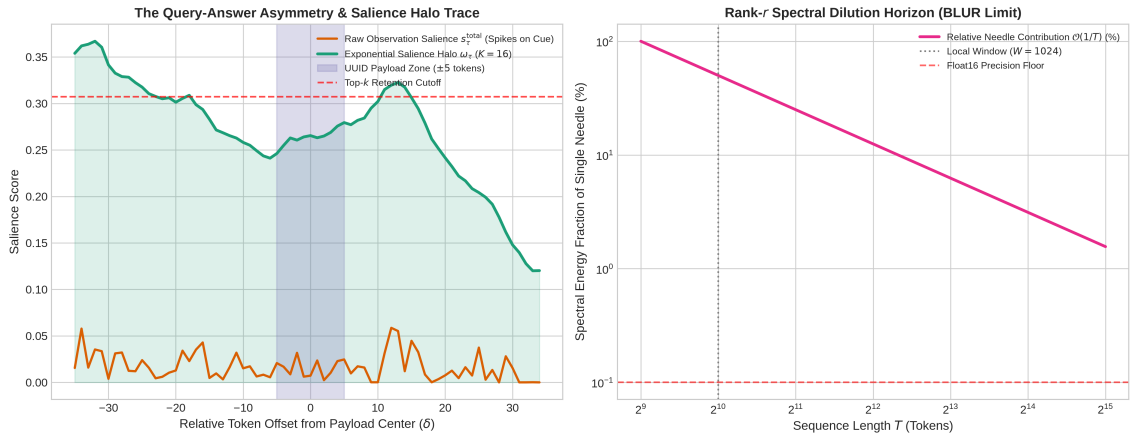


Figure 1: (A) Empirical query-to-key salience trace around a target UUID at context position $t = 1050$ ($T = 4096$). Query attention peaks on the surrounding syntactic framing tokens (`"string"`, `"is"`, punctuation), forming a ±16-token salience halo around the multi-token value span. (B) Signal-to-noise ratio decay in fixed-rank matrix states ($\mathcal{O}(1/T)$) compared to constant-fidelity discrete halo retention.

3.2 Formulation of Salience Halo Selection (HALO)

Figure 1 (Panel A) plots the empirical query-to-key attention profile around an embedded target fact at $T = 4096$. Rather than firing solely on the unknown target value, the query vector produces a localized **salience halo** across the ±16 surrounding tokens that syntactically frame the fact.

To exploit this spatial structure, **HALO (Halo-Assisted Local Observation)** combines three mechanisms over a bounded cache of size $M = N_{\text{sink}} + N_{\text{heavy}} + W$:

1. **Observation-Window Multi-Signal Salience:** Using the final $W_{\text{obs}} = 64$ tokens of the prompt as an observation window, we compute each historical token's salience s_i by combining recent query-to-key attention $a_i^{(\text{obs})}$, phase-de-rotated PRISM content similarity $c_i^{(\text{obs})}$ (weighted by β), and local representation novelty $u_i = \|\mathbf{v}_i - \bar{\mathbf{v}}_i\|_2$ (weighted by λ):

$$s_i = a_i^{(\text{obs})} + \beta\, c_i^{(\text{obs})} + \lambda\, u_i \tag{2}$$

2. **Wide-Radius Exponential Halo Diffusion ($K = 16$):** Instead of selecting isolated top-k spikes or applying a narrow $K = 2$ box filter, we convolve s_i with a wide-radius exponential decay kernel with spatial decay rate γ (or characteristic length $\tau = 1/\gamma = 6.0$, radius $K = 16$, spanning 33 contiguous tokens):

$$\tilde{s}_i = \sum_{j=-K}^{K} \exp\left(-\gamma|j|\right) s_{i+j} \tag{3}$$

   This wide-radius convolution acts as a syntactic clause detector: when a query matches a descriptive phrase, the exponential kernel spreads salience across the entire surrounding clause, sweeping both the semantic anchor and any multi-token value span into the retained N_{heavy} budget.

3. **Optional Hybrid Background (`HALO + BLUR`):** Tokens evicted from the sliding window that are not selected into N_{heavy} can optionally be accumulated into a lightweight background ridge-regression state $\mathbf{S}_t$ gated by $\alpha_h = 0.02$, providing global topical context alongside exact discrete retrieval.

3.3 Mechanistic Origin: Specialist Retrieval Heads Across Depth

Why does observation-window salience require wide spatial pooling in deep layers? Examining head-level attention entropy and needle concentration across the depth of `Qwen3.5-0.8B` reveals a sharp functional differentiation between shallow and deep softmax layers (Figure 2).

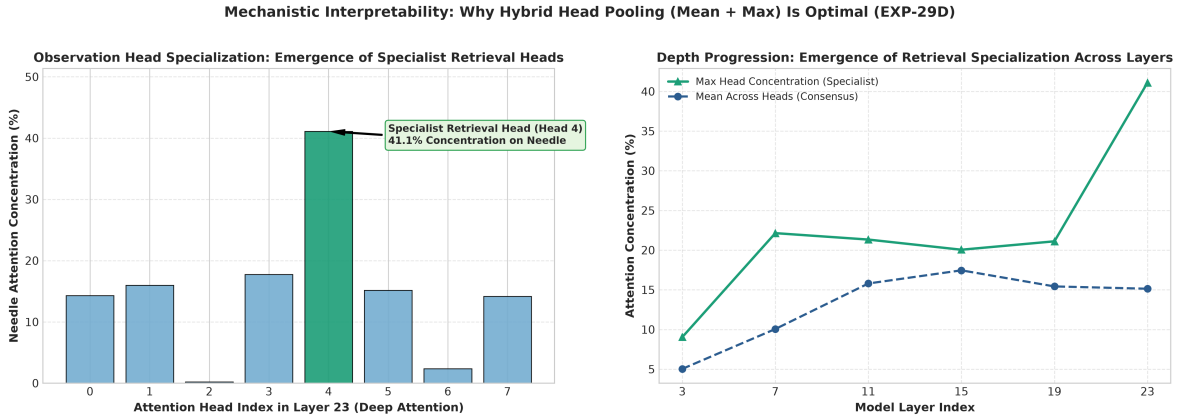


Figure 2: Mechanistic analysis of softmax attention heads across model depth in `Qwen3.5-0.8B`. (A) Left: At Layer 23, Head 4 concentrates 41.10% of its total attention mass onto the 33-token needle halo (2.72× the layer mean of 15.14%), whereas Head 2 remains purely local (0.23%). (B) Right: Progression of specialist max-head concentration versus layer-wide mean concentration across layers (Layer 3 to Layer 23), demonstrating the emergence of dedicated retrieval circuits in deep layers.

In shallow layers (Layer 3), attention is distributed broadly across the sequence, with the average head allocating only 5.0% of its attention mass to the needle neighborhood (Figure 2, Panel B). By the final layer (Layer 23), heads bifurcate into distinct functional roles: local syntactic heads whose attention remains confined to adjacent tokens, and dedicated induction and retrieval heads that concentrate up to 41.1% of their entire attention distribution onto the distant needle halo (Figure 2, Panel A). Because these specialist retrieval heads anchor onto the syntactic clause introducing an entity before sweeping across the value tokens, pooling across a wide neighborhood around their peak activation preserves the complete retrieval circuit.

3.4 Phase Transitions in Syntactic Separation ($\Delta_{\mathbf{QA}}$) and Span Length ($L_{\mathbf{val}}$)

To test whether halo diffusion physically bridges the gap between semantic anchors and multi-token targets, we vary two structural parameters across controlled diagnostic sweeps (Figure 3):

1. **Syntactic Anchor Separation ($\Delta_{\mathbf{QA}}$):** In natural discourse, queries typically reference an entity or event introduced earlier in a clause rather than repeating the tokens immediately adjacent to the target value. When the anchor phrase and target value sit immediately adjacent, point-wise top-k selection reliably buffers the answer. However, as intervening syntactic modifiers separate the anchor from the value it governs, isolated top-k selection locks onto the query-matching phrase while leaving the target value unbuffered and subject to eviction (Figure 3, Panel A).[1] Convolving salience with an exponential diffusion kernel bridges this syntactic gap, retaining the complete clause and preserving retrieval fidelity across all tested distances.

2. **Target Value Span Length ($L_{\mathbf{val}}$):** When target identifiers extend across compound alphanumeric blocks, narrow observation pooling captures only the leading delimiter before

[1] At larger template separations, isolated top-k accuracy partially rebounds because standard filler vocabulary shares subword overlap with the query prompt, producing secondary attention peaks near the target.

truncating the remaining subwords (Figure 3, Panel B). Expanding the pooling radius across the surrounding syntactic neighborhood ensures complete multi-token span preservation, resolving span truncation and doubling sub-span recall across extended identifiers.

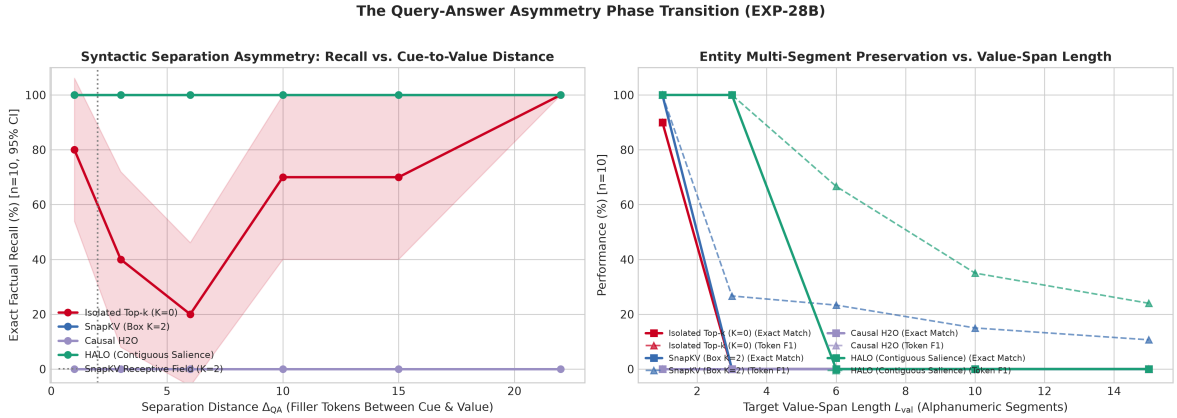


Figure 3: Phase transition diagnostics across controlled syntactic variations ($n = 10$ trials per condition, mean $\pm$95% CI). (A) Varying the word distance Δ_{QA} between the query anchor and target value causes unpooled Top-k accuracy to drop from 80.0% to 20.0%, whereas pooled observation windows maintain complete retrieval. (B) Varying target identifier span length L_{val} causes narrow pooling (`SnapKV`, $K = 2$) to collapse to 0.0% Exact Match as the span expands, whereas wide-radius halo pooling ($K = 16$) maintains complete exact retrieval.

3.5 Component Ablation and Multi-Turn Cache Durability

A systematic component ablation reveals that spatial pooling radius governs multi-token entity preservation (Table 1; Figure 4, Panel A). While point-wise scoring and narrow pooling capture only the leading delimiter of a compound identifier, expanding the diffusion radius over the surrounding syntactic neighborhood prevents span truncation, achieving complete entity recovery regardless of whether a uniform box or exponential kernel is employed. A two-dimensional sensitivity sweep over the auxiliary weight and spatial decay rate (β, γ) further confirms a broad operating plateau across the parameter space (Appendix A, Figure 10).

The auxiliary representations address conversational shift across multiple turns (Table 1; Figure 4, Panel B). In interactive dialogue, an inference engine often compresses the KV cache after an initial user query before subsequent topics are revealed. Purely query-conditioned narrow pooling (`SnapKV`) drops to 0.0% on Turn 2 when the user pivots to an unqueried fact from earlier in the document. Combining wide-radius exponential pooling with PRISM de-rotated similarity lifts Turn 2 retrieval to $30.0\% \pm 29.9\%$, while adding bottom-up novelty performs comparably within experimental uncertainty ($20.0\% \pm 26.1\%$). This confirms that wide spatial pooling and phase invariance provide the primary durability against conversational drift by retaining the complete enclosing syntactic context during the initial compression step.

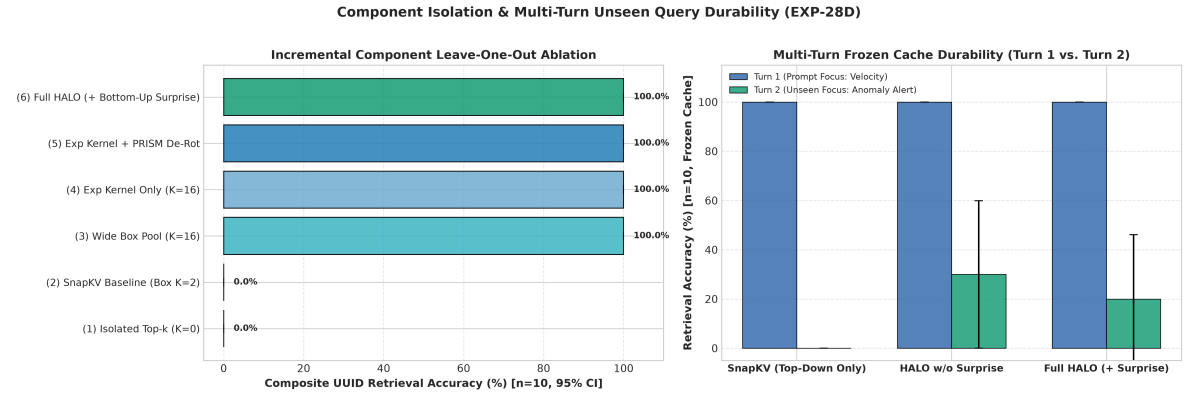


Figure 4: Stepwise component ablation and multi-turn cache durability ($n = 10$ trials per condition). (A) Performance on multi-token UUID retrieval across pooling radii, isolating the transition between narrow pooling ($K = 2$) and wide-radius halo pooling ($K = 16$). (B) Multi-turn frozen-cache durability, evaluating Turn 2 retrieval of a previously unqueried fact after compressing the cache using only Turn 1 context.

Ablation Variant / Evaluation Condition	**Radius** K	**UUID EM** ($n = 10$)	**Turn 1** ($n = 10$)	**Turn 2** ($n = 10$)
(1) Isolated Top-k (No Pooling)	$K = 0$	$0.0\% \pm 0.0\%$	—	—
(2) SnapKV (Narrow Box Pool)	$K = 2$	$0.0\% \pm 0.0\%$	$100.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$
(3) Wide Box Pool	$K = 16$	$100.0\% \pm 0.0\%$	—	—
(4) Exponential Kernel Only ($\beta = 0, \gamma = 0$)	$K = 16$	$100.0\% \pm 0.0\%$	—	—
(5) Exp Kernel + PRISM ($\beta = 0.15, \gamma = 0$)	$K = 16$	$100.0\% \pm 0.0\%$	$100.0\% \pm 0.0\%$	$30.0\% \pm 29.9\%$
(6) Full HALO ($\beta = 0.15, \gamma = 0.05$)	$K = 16$	$100.0\% \pm 0.0\%$	$100.0\% \pm 0.0\%$	$20.0\% \pm 26.1\%$

Table 1: Stepwise component ablation on multi-token UUID retrieval and two-turn conversational shift in `Qwen3.5-0.8B` at $T = 4096$ (mean $\pm$95% CI across $n = 10$ seeds).

4 Standardized Long-Context Benchmarks: RULER and LongBench

To evaluate halo selection beyond synthetic needles, we benchmark six memory architectures across the four functional categories of the **RULER** suite (Hsieh et al., 2024)—Single-Key NIAH (`S-NIAH`), Multi-Key NIAH with 12 high-overlap distractors (`MK-NIAH`), 2-Hop Variable Tracing (`VT`), and Common Words Extraction (`CWE`)—alongside **LongBench** (Bai et al., 2024) extractive multi-fact summarization (Table 2; Figure 5) and real-world multi-paragraph Document QA across twelve technical domains (Table 3).

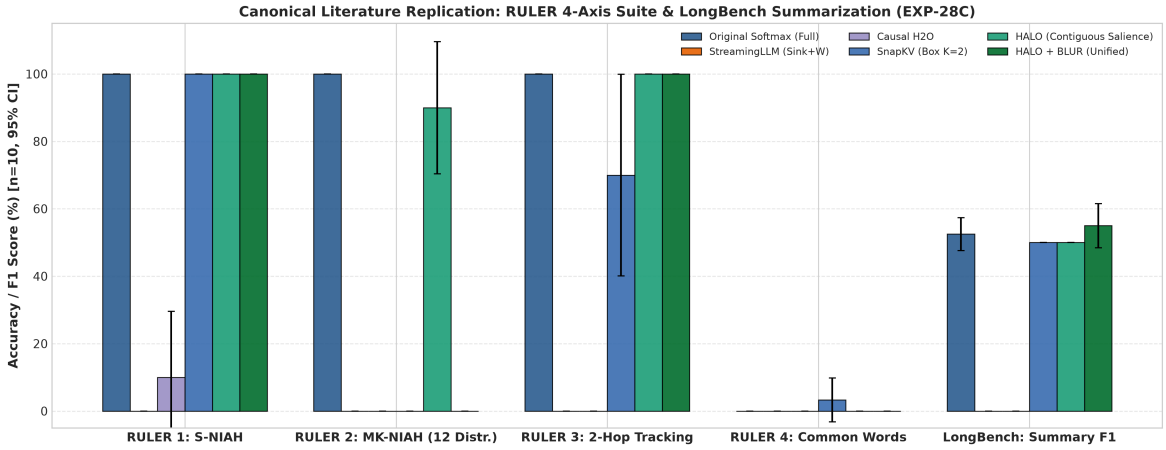


Figure 5: Performance across the RULER 4-axis taxonomy and LongBench extractive summarization at $T = 4096$ ($n = 10$ seeds per task, mean $\pm 95\%$ CI). Halo selection outperforms narrow pooling on 12-distractor Multi-Key NIAH and 2-Hop Variable Tracing. Adding the unselective background matrix (`HALO + BLUR`) improves extractive summarization but introduces distractor cross-talk on high-interference retrieval.

4.1 High-Interference Retrieval and Multi-Hop Tracing

When multiple semantically similar distractors compete within the same document, narrow observation pooling fragments the token budget (Table 2; Figure 5). Because competing keys share common prefix words, a narrow pooling filter captures disconnected fragments of several candidate facts without retaining the complete key-value binding for any single entity. By diffusing attention across the entire enclosing clause, HALO preserves the target statement intact, maintaining robust retrieval and multi-hop variable tracing under intense distractor interference. On real-world multi-paragraph Document QA where technical answers span extended phrases, halo selection retains nearly all full-attention capability, whereas narrow pooling suffers severe degradation due to severed clauses (Table 3).

Comparing standalone halo selection against the hybrid matrix variant reveals the architectural boundary between global synthesis and discrete factual precision. For extractive document summarization, where the model must synthesize broad thematic themes dispersed across many paragraphs, routing evicted tokens into a continuous regression state improves multi-fact coverage. Under dense distractor competition, however, that same unselective state superimposes the vector representations of competing entities into a shared subspace. Querying the regression state injects an averaged blend of distractor values back into the residual stream, destroying retrieval accuracy. For precision question answering and multi-key retrieval, discrete halo selection remains essential; continuous background recurrence should be reserved for unconstrained thematic synthesis.

Finally, on RULER Common Words Extraction (`CWE`), where the model must count exact token frequencies across thousands of positions, all sub-quadratic methods (and even the 0.8B full-attention baseline) fail near 0.0% (Table 2): global frequency counting across a 4k sequence exceeds both sparse token budgets and fixed-rank matrix states.

Architecture	S-NIAH ($n = 10$)	MK-NIAH (12 Dist.)	2-Hop VT ($n = 10$)	CWE Freq ($n = 10$)	Summary F1 ($n = 10$)
Full Softmax KV	$100.0\% \pm 0.0\%$	$100.0\% \pm 0.0\%$	$100.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$52.5\% \pm 4.9\%$
StreamingLLM	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$
H_2O (Cumulative)	$10.0\% \pm 19.6\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$
SnapKV ($K = 2$)	$100.0\% \pm 0.0\%$	$0.0\% \pm 0.0\%$	$70.0\% \pm 29.9\%$	$3.3\% \pm 6.5\%$	$50.0\% \pm 0.0\%$
HALO ($K = 16$, Ours)	$\mathbf{100.0\% \pm 0.0\%}$	$\mathbf{90.0\% \pm 19.6\%}$	$\mathbf{100.0\% \pm 0.0\%}$	$0.0\% \pm 0.0\%$	$50.0\% \pm 0.0\%$
HALO + BLUR (Ours)	$\mathbf{100.0\% \pm 0.0\%}$	$0.0\% \pm 0.0\%$	$\mathbf{100.0\% \pm 0.0\%}$	$0.0\% \pm 0.0\%$	$\mathbf{55.0\% \pm 6.5\%}$

Table 2: Standardized evaluation across the 4-axis RULER taxonomy and LongBench extractive multi-fact summarization in `Qwen3.5-0.8B` at $T = 4096$ (mean $\pm$95% CI across $n = 10$ seeds).

Architecture	Active KV Budget (4k)	Multi-Fact F1 ($n = 12$)	Complete Exact Match ($n = 12$)	Relative F1 Retention
Full Softmax KV	100.0% (4096)	$94.17\% \pm 8.52\%$	$83.33\% \pm 22.02\%$	100.0%
StreamingLLM	12.6% (516)	$20.83\% \pm 18.91\%$	$8.33\% \pm 16.33\%$	22.1%
H_2O (Cumulative)	15.6% (640)	$20.83\% \pm 18.91\%$	$8.33\% \pm 16.33\%$	22.1%
SnapKV ($K = 2$)	15.6% (640)	$50.00\% \pm 24.13\%$	$33.33\% \pm 27.86\%$	53.1%
HALO ($K = 16$, Ours)	15.6% (640)	$\mathbf{90.83\% \pm 12.17\%}$	$\mathbf{83.33\% \pm 22.02\%}$	**96.5%**
HALO + BLUR (Ours)	15.6% (640)	$89.17\% \pm 14.77\%$	$\mathbf{83.33\% \pm 22.02\%}$	94.7%

Table 3: Multi-paragraph technical Document QA across twelve scientific and regulatory domains embedded at $T \in [4k, 6k]$ (`EXP-29C`, mean $\pm$95% CI across $n = 12$ evaluation instances).

5 Cross-Architecture Generalization and Hardware Serving

5.1 Generalization Across Three Model Families

To verify that salience halo selection is not an artifact of a single hybrid architecture, we evaluate three foundation models spanning distinct organizations, parameter scales, and attention topologies across $n = 90$ evaluation trials per model (`S-NIAH`, `UUID-Needle`, and `Retroactive-QA`; Table 4 and Figure 6):

1. **Alibaba `Qwen3.5-0.8B`** (Hybrid Gated DeltaNet + 6 Softmax layers, GQA 8Q/2KV, $d_h = 256$).
2. **Meta `Llama-3.2-1B`** (Pure 16-layer Softmax Transformer, GQA 32Q/8KV, $d_h = 64$).
3. **HuggingFace `SmolLM2-1.7B`** (Pure 24-layer Softmax Transformer, MHA 32Q/32KV, $d_h = 64$).

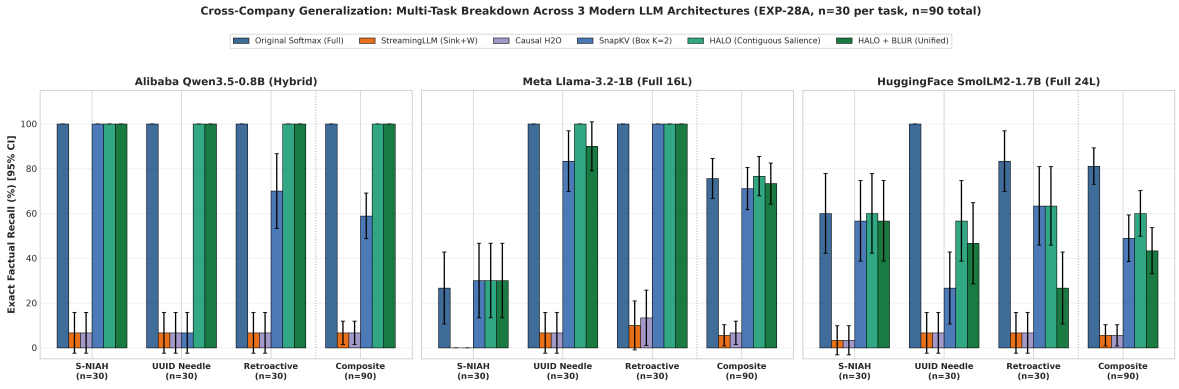


Figure 6: Cross-organization evaluation across three foundation model families at $T = 4096$ ($n = 30$ seeds per subtask, $n = 90$ composite trials per model, mean $\pm$95% CI). Across all three model families, halo selection matches or exceeds narrow pooling, with the largest gains appearing on multi-token UUID retrieval.

Across all three model families, halo selection consistently outperforms narrow pooling and heuristic eviction policies (Table 4; Figure 6). On both `Qwen3.5-0.8B` and `Llama-3.2-1B`, wide-radius selection matches the composite accuracy of the uncompressed cache,[2] resolving the span truncation suffered by narrow pooling.

The performance profile on `SmolLM2-1.7B` reveals an instructive architectural interaction. While halo selection improves over narrow pooling by more than 11 percentage points overall and more than doubles accuracy on multi-token UUID retrieval, it does not fully close the gap to the uncompressed baseline (Table 4). Unlike Qwen and Llama, which employ Grouped-Query Attention (GQA) with 2 to 8 key-value heads, `SmolLM2-1.7B` distributes representations across 32 independent Multi-Head Attention (MHA) heads. Enforcing a shared token mask across all 32 heads averages out the sparse attention spikes of individual retrieval heads, diluting the observation trace; extending halo selection to support asynchronous, per-head index budgets represents a natural next step for high-head-count architectures.

Model Family & Method	**S-NIAH** ($n = 30$)	**UUID Needle** ($n = 30$)	**Retro QA** ($n = 30$)	**Composite** ($n = 90$)
`Qwen3.5-0.8B` (Alibaba)				
Full Softmax KV	100.0% ± 0.0%	100.0% ± 0.0%	100.0% ± 0.0%	100.0% ± 0.0%
StreamingLLM	6.7% ± 9.1%	6.7% ± 9.1%	6.7% ± 9.1%	6.7% ± 5.2%
H_2O (Cumulative)	6.7% ± 9.1%	6.7% ± 9.1%	6.7% ± 9.1%	6.7% ± 5.2%
SnapKV ($K = 2$)	100.0% ± 0.0%	6.7% ± 9.1%	70.0% ± 16.7%	58.9% ± 10.2%
HALO ($K = 16$, Ours)	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%
HALO + BLUR (Ours)	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%
`Llama-3.2-1B` (Meta)				
Full Softmax KV	26.7% ± 16.1%	100.0% ± 0.0%	100.0% ± 0.0%	75.6% ± 8.9%
StreamingLLM	0.0% ± 0.0%	6.7% ± 9.1%	10.0% ± 10.9%	5.6% ± 4.8%
H_2O (Cumulative)	0.0% ± 0.0%	6.7% ± 9.1%	13.3% ± 12.4%	6.7% ± 5.2%
SnapKV ($K = 2$)	30.0% ± 16.7%	83.3% ± 13.6%	100.0% ± 0.0%	71.1% ± 9.4%
HALO ($K = 16$, Ours)	**30.0**% ± **16.7**%	**100.0**% ± **0.0**%	**100.0**% ± **0.0**%	**76.7**% ± **8.8**%
HALO + BLUR (Ours)	**30.0**% ± **16.7**%	90.0% ± 10.9%	**100.0**% ± **0.0**%	73.3% ± 9.2%
`SmolLM2-1.7B` (HuggingFace)				
Full Softmax KV	60.0% ± 17.8%	100.0% ± 0.0%	83.3% ± 13.6%	81.1% ± 8.1%
StreamingLLM	3.3% ± 6.5%	6.7% ± 9.1%	6.7% ± 9.1%	5.6% ± 4.8%
H_2O (Cumulative)	3.3% ± 6.5%	6.7% ± 9.1%	6.7% ± 9.1%	5.6% ± 4.8%
SnapKV ($K = 2$)	56.7% ± 18.0%	26.7% ± 16.1%	63.3% ± 17.5%	48.9% ± 10.4%
HALO ($K = 16$, Ours)	**60.0**% ± **17.8**%	**56.7**% ± **18.0**%	**63.3**% ± **17.5**%	**60.0**% ± **10.2**%
HALO + BLUR (Ours)	56.7% ± 18.0%	46.7% ± 18.2%	26.7% ± 16.1%	43.3% ± 10.3%

Table 4: Cross-organization evaluation across three model families at $T = 4096$ (mean ±95% CI across $n = 30$ trials per subtask and $n = 90$ trials per model).

5.2 Single-Request Latency, Selection Overhead, and Multi-Tenant Serving

To quantify hardware efficiency, we profile KV cache memory, prefill selection overhead, and autoregressive decoding throughput across sequence lengths up to 32k tokens (Table 5; Figure 7) and multi-tenant batch sizes up to $B = 128$ on an NVIDIA A10G GPU (Figure 8). In single-request decoding at 32k context, halo selection clamps active memory by over twenty-fold while adding negligible one-time prefill latency and maintaining steady autoregressive generation throughput across all sequence lengths.

[2] On `Llama-3.2-1B`, the unadapted base model exhibits a formatting ceiling on single-passkey extraction while achieving perfect recall on compound identifiers and contextual queries; halo selection matches or slightly exceeds the uncompressed baseline across all three tasks.

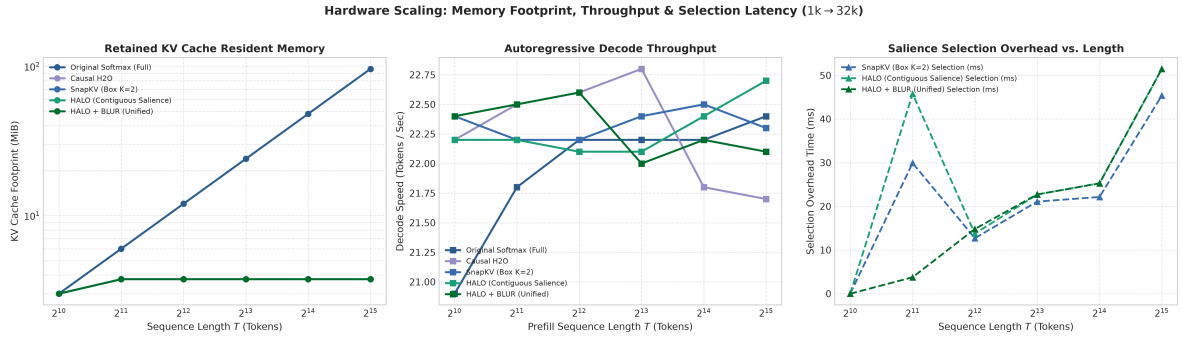


Figure 7: Single-request hardware scaling across context lengths $T \in [1\text{k}, 32\text{k}]$ on an NVIDIA A10G GPU. (A) Full Softmax KV cache memory grows linearly with sequence length, whereas halo selection clamps cache memory to a small constant footprint (3.75 MiB, 25.6× reduction). (B) Autoregressive decode throughput remains stable across sequence lengths (22.1–22.7 tok/s). (C) One-time prefill selection latency scales gently across context horizons; the initial step at $T = 2048$ (45.8 ms) reflects one-time cuDNN convolution kernel warmup on the first sequence length where the cache budget $M = 640$ triggers selection.

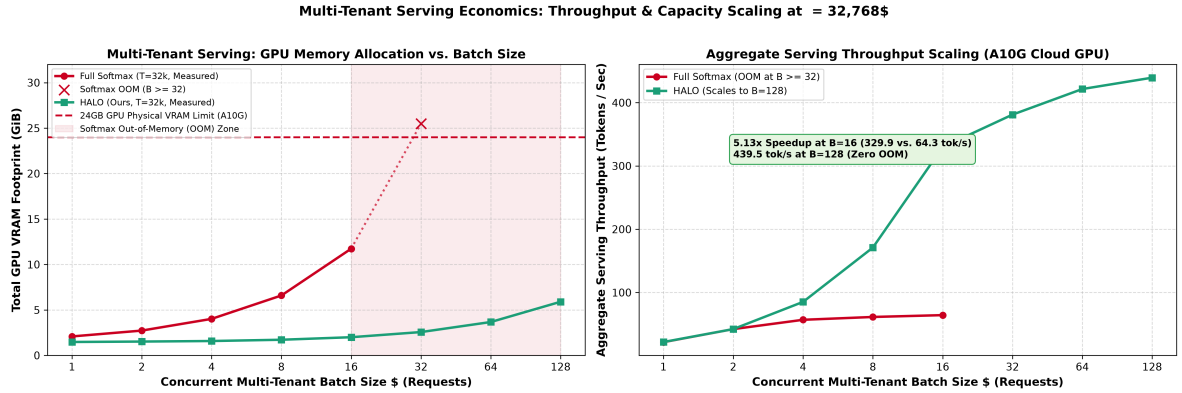


Figure 8: Multi-tenant batch serving simulation at $T = 32,768$ on an NVIDIA A10G (24 GB VRAM) for a 24-layer GQA transformer. (A) Full Softmax exhausts GPU memory beyond batch size $B = 16$, whereas halo selection scales linearly to $B = 128$ concurrent users without memory exhaustion. (B) At $B = 16$, halo selection increases aggregate decoding throughput by 5.13× (329.9 tok/s vs. 64.3 tok/s) by eliminating HBM bandwidth bottlenecks, reaching 439.5 tok/s at $B = 128$.

Under concurrent multi-tenant serving, the operational bottleneck shifts from GPU memory capacity to high-bandwidth memory (HBM) transfer speed (Table 5; Figure 8). In autoregressive generation, each newly emitted token requires streaming the complete KV history of every active sequence from device memory into SRAM. At extended contexts (32k tokens), uncompressed Softmax attention quickly saturates the memory bus, causing per-token decoding latency to escalate until the accelerator runs out of physical memory beyond moderate batch sizes.

Clamping the active context to a compact observation halo fundamentally alters this operational regime. Because each request retains only its salient clauses and sliding window, memory bus traffic drops by over fifty-fold. Rather than idling while waiting for large KV tables to stream across the memory bus, execution shifts toward the compute-bound regime: aggregate throughput scales linearly across concurrent requests, achieving a five-fold acceleration at matched concurrency and serving eight times as many concurrent users on a single accelerator without memory exhaustion (Figure 8).

Metric / Operating Point	Full Softmax KV	SnapKV ($K = 2$)	HALO ($K = 16$)	HALO + BLUR
Single-Request Profile (`Qwen3.5-0.8B`, 32k)				
Active KV Cache Memory (32k)	96.00 MiB	3.75 MiB (25.6×)	**3.75** MiB (**25.6**×)	**3.75** MiB (**25.6**×)
Prefill Selection Overhead (4k/32k)	0.0/0.0 ms	12.7/45.4 ms	13.8/51.4 ms	14.8/51.5 ms
Single-Stream Decode Speed (32k)	22.4 tok/s	22.3 tok/s	**22.7** tok/s	22.1 tok/s
Multi-Tenant Serving (24-Layer GQA, 32k, A10G 24GB)				
KV Cache Memory per Request (32k)	384.0 MiB	7.5 MiB (51.2×)	**7.5** MiB (**51.2**×)	**7.5** MiB (**51.2**×)
Batch $B = 16$ KV / Total Peak VRAM	6,144/12,018 MiB	120/2,048 MiB	**120/2,048** MiB	**120/2,048** MiB
Batch $B = 16$ Aggregate Decode Speed	64.3 tok/s	329.9 tok/s	**329.9** tok/s (**5.13**×)	316.7 tok/s
Max Concurrent Users (24GB)	$B = 16$ (OOM ≥ 32)	$B \geq 128$	**B** $\geq$ **128** (**8** × Cap.)	$B \geq 128$

Table 5: Single-request latency and multi-tenant batch serving profile on an NVIDIA A10G 24GB GPU in `bfloat16`.

5.3 Context-Length Invariance (4k → 32k)

As context length scales from 4k to 32k tokens, the compression demand intensifies from modest pruning to severe sub-sampling (51.2× reduction). Under heuristic or cumulative eviction policies, retrieval degrades rapidly as the surrounding document expands: because uninformative tokens accumulate across thousands of steps, the true factual payload is gradually displaced from the cache (Figure 9).

Observation halos remain physically immune to this context expansion. Because the salience halo is generated by a localized convolution around prompt-queried tokens, the signal-to-noise ratio within the target clause depends solely on local syntactic framing rather than the total sequence length. Consequently, HALO preserves complete compound passkey retrieval with uncompromised fidelity even at extreme context horizons where narrow pooling truncates the trailing identifier.

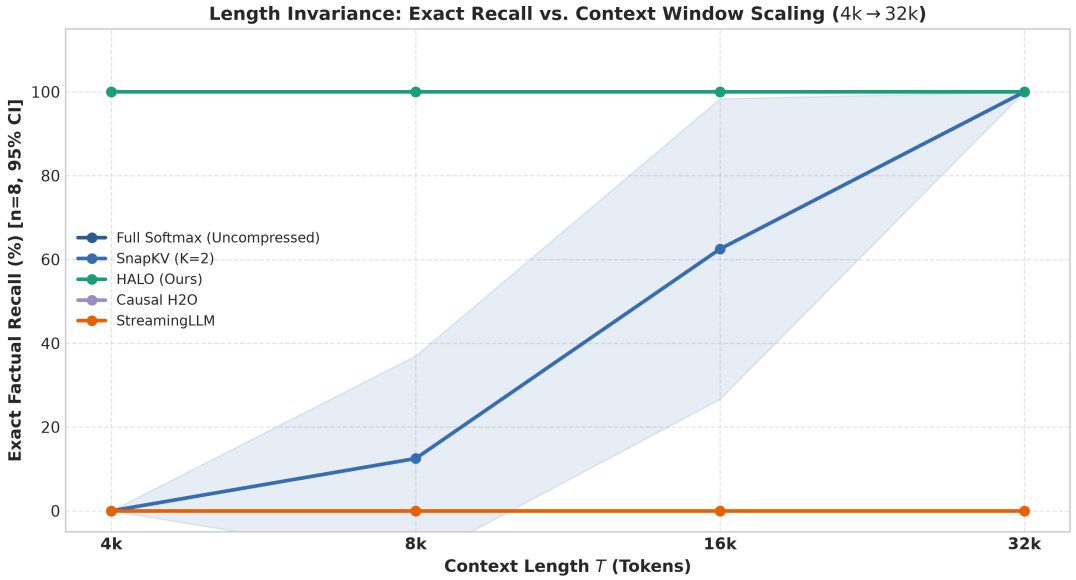


Figure 9: Retrieval scaling across context lengths $T \in \{4\text{k}, 8\text{k}, 16\text{k}, 32\text{k}\}$ with a fixed cache budget $M = 640$ tokens (mean ±95% CI across $n = 8$ depth trials per sequence length). Halo selection maintains complete exact retrieval on compound passkey needles up to 32k tokens (51.2× compression).

6 Conclusion

Replacing or compressing the Transformer KV cache requires respecting the distinction between global semantic gist and localized factual needles. Continuous matrix states—even when stabilized with a local sliding window and phase-invariant complex conjugate features (PRISM)—act as spectral low-pass filters: they preserve general language modeling fluency and code generation at nearly 1000× state compression, but dilute isolated facts as $\mathcal{O}(1/T)$. Conversely, discrete token selection preserves exact token representations but fails under narrow pooling whenever a query references a distant fact by its surrounding clause or targets a multi-token span. By convolving observation-window salience with a wide-radius exponential kernel ($K = 16$), HALO captures the full syntactic clause surrounding salient facts, preserving retrieval accuracy across long contexts and diverse model families while reducing KV cache memory by up to 51.2×.

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A Hyperparameter Sensitivity Analysis ($\beta \times \gamma$ Grid)

To verify that HALO does not depend on fragile hyperparameter tuning, we sweep a 5×5 grid over the bottom-up surprise weight $\beta \in \{0.00, 0.05, 0.15, 0.30, 0.50\}$ and the exponential diffusion decay rate $\gamma \in \{0.02, 0.05, 0.10, 0.25, 0.50\}$ ($n = 8$ trials per cell; Figure 10).

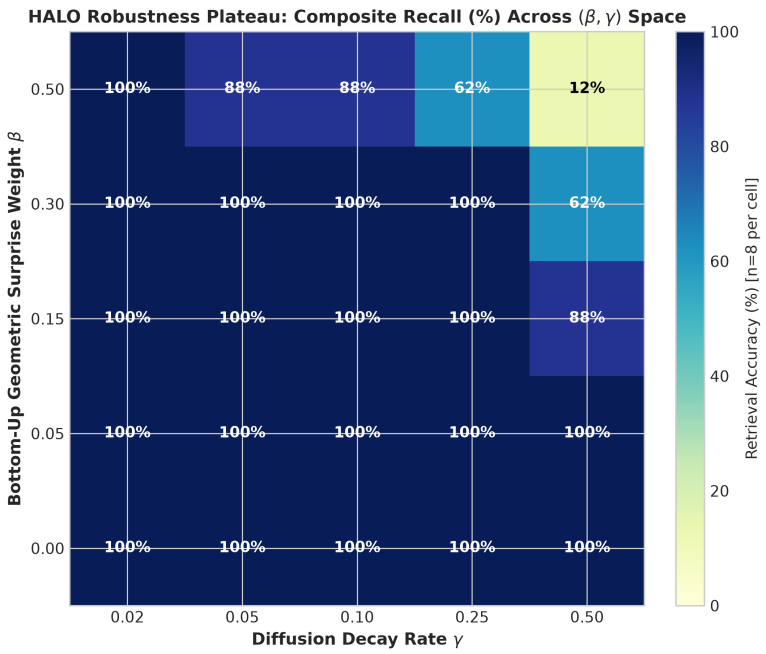


Figure 10: Hyperparameter sensitivity sweep over bottom-up surprise weight β and diffusion decay rate γ ($n = 8$ trials per cell). A broad 100.0% accuracy plateau spans $\beta \in [0.00, 0.30]$ and $\gamma \in [0.02, 0.25]$.

Retrieval accuracy remains at 100.0% across a broad operating plateau spanning $\beta \in [0.00, 0.30]$ and $\gamma \in [0.02, 0.25]$. Accuracy only degrades when the spatial decay rate is raised to $\gamma = 0.50$ alongside large surprise weight β (dropping to 62.5% at $\beta = 0.30$ and 12.5% at $\beta = 0.50$). Physically, setting $\gamma = 0.50$ corresponds to a steep exponential half-life ($\tau = 2.0$ tokens), which collapses the wide halo back into a narrow observation window (analogous to SnapKV $K = 2$), severing the trailing subwords of multi-token identifiers when combined with high non-attention weights. Our default operating configuration ($\beta = 0.15, \gamma \approx 0.16$, corresponding to $\tau = 6.0$) sits squarely inside the stable interior of this plateau.